ALLIANCE UNIVERSITY | NAAC GRADE A+ ACCREDITED UNIVERSITY

Database Management System

Assignment 5

CAB BOOKING SYSTEM

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Course	Database Management System
Programme	BCA AI/ML
Semester	III

Creating the database

```
CREATE DATABASE CabBookingDB;
USE CabBookingDB;
```

Creating the tables

```
CREATE TABLE Customer (
    Customer_ID    VARCHAR(10)  NOT NULL,
    Customer_Name  VARCHAR(50)  NOT NULL,
    Phone          VARCHAR(15),
    Email          VARCHAR(50),
    Address        VARCHAR(100),
    PRIMARY KEY (Customer_ID)
);

CREATE TABLE Driver (
    Driver_ID      VARCHAR(10)  NOT NULL,
    Driver_Name    VARCHAR(50)  NOT NULL,
    Phone          VARCHAR(15),
    License_No     VARCHAR(20),
    Availability   VARCHAR(15),
    PRIMARY KEY (Driver_ID)
);

CREATE TABLE Car (
    Car_ID         VARCHAR(10)  NOT NULL,
    Car_Number     VARCHAR(15)  NOT NULL,
    Car_Model      VARCHAR(30),
    Car_Type       VARCHAR(20),
    Fare_Per_KM    DECIMAL(6,2),
    Driver_ID      VARCHAR(10),
    PRIMARY KEY (Car_ID),
    FOREIGN KEY (Driver_ID) REFERENCES Driver(Driver_ID)
);

CREATE TABLE Booking (
    Booking_ID       VARCHAR(10) NOT NULL,
    Customer_ID      VARCHAR(10) NOT NULL,
    Car_ID           VARCHAR(10) NOT NULL,
    Booking_Date     DATE,
    Pickup_Location  VARCHAR(50),
    Drop_Location    VARCHAR(50),
    Distance_KM      INT,
    Booking_Status   VARCHAR(15),
    PRIMARY KEY (Booking_ID),
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID),
    FOREIGN KEY (Car_ID)      REFERENCES Car(Car_ID)
);

CREATE TABLE Payment (
    Payment_ID      VARCHAR(10) NOT NULL,
    Booking_ID      VARCHAR(10) NOT NULL,
    Amount          DECIMAL(8,2),
    Payment_Date    DATE,
    Payment_Method  VARCHAR(10),
    Payment_Status  VARCHAR(10),
    PRIMARY KEY (Payment_ID),
    FOREIGN KEY (Booking_ID) REFERENCES Booking(Booking_ID)
);
```

Verifying the tables

```
SHOW TABLES;
```

Output:

Tables_in_cabbookingdb
customer
driver
car
booking
payment

```
DESCRIBE Customer;
```

Output:

Field	Type	Null	Key	Default
Customer_ID	varchar(10)	NO	PRI	NULL
Customer_Name	varchar(50)	NO		NULL
Phone	varchar(15)	YES		NULL
Email	varchar(50)	YES		NULL
Address	varchar(100)	YES		NULL

```
DESCRIBE Driver;
```

Output:

Field	Type	Null	Key	Default
Driver_ID	varchar(10)	NO	PRI	NULL
Driver_Name	varchar(50)	NO		NULL
Phone	varchar(15)	YES		NULL
License_No	varchar(20)	YES		NULL
Availability	varchar(15)	YES		NULL

```
DESCRIBE Car;
```

Output:

Field	Type	Null	Key	Default
Car_ID	varchar(10)	NO	PRI	NULL
Car_Number	varchar(15)	NO		NULL
Car_Model	varchar(30)	YES		NULL
Car_Type	varchar(20)	YES		NULL
Fare_Per_KM	decimal(6,2)	YES		NULL
Driver_ID	varchar(10)	YES	MUL	NULL

```
DESCRIBE Booking;
```

Output:

Field	Type	Null	Key	Default
Booking_ID	varchar(10)	NO	PRI	NULL
Customer_ID	varchar(10)	NO	MUL	NULL
Car_ID	varchar(10)	NO	MUL	NULL
Booking_Date	date	YES		NULL

Field	Type	Null	Key	Default
Pickup_Location	varchar(50)	YES		NULL
Drop_Location	varchar(50)	YES		NULL
Distance_KM	int	YES		NULL
Booking_Status	varchar(15)	YES		NULL

```
DESCRIBE Payment;
```

Output:

Field	Type	Null	Key	Default
Payment_ID	varchar(10)	NO	PRI	NULL
Booking_ID	varchar(10)	NO	MUL	NULL
Amount	decimal(8,2)	YES		NULL
Payment_Date	date	YES		NULL
Payment_Method	varchar(10)	YES		NULL
Payment_Status	varchar(10)	YES		NULL

Checking the relationships

```
SELECT TABLE_NAME, COLUMN_NAME, REFERENCED_TABLE_NAME, REFERENCED_COLUMN_NAME
FROM INFORMATION_SCHEMA.KEY_COLUMN_USAGE
WHERE TABLE_SCHEMA = 'CabBookingDB'
  AND REFERENCED_TABLE_NAME IS NOT NULL;
```

Output:

TABLE_NAME	COLUMN_NAME	REFERENCED_TABLE_NAME	REFERENCED_COLUMN_NAME
Car	Driver_ID	Driver	Driver_ID
Booking	Car_ID	Car	Car_ID
Booking	Customer_ID	Customer	Customer_ID
Payment	Booking_ID	Booking	Booking_ID

Foreign key	Relationship it represents
Car.Driver_ID → Driver	DRIVER is assigned to CAR (1:N)
Booking.Customer_ID → Customer	CUSTOMER makes BOOKING (1:N)
Booking.Car_ID → Car	CAR is used for BOOKING (1:N)
Payment.Booking_ID → Booking	BOOKING has PAYMENT (1:1, made strict in Assignment 6 with UNIQUE)